Multichannel scheduling and routing topologies for throughput-delay tradeoff

Joint frequency and time slot assignment

- NP-hard for general graphs
- Approximation algorithms with constant factor and logarithmic bounds
- Bounded-degree, minimum-radius spanning trees (BDMRST)

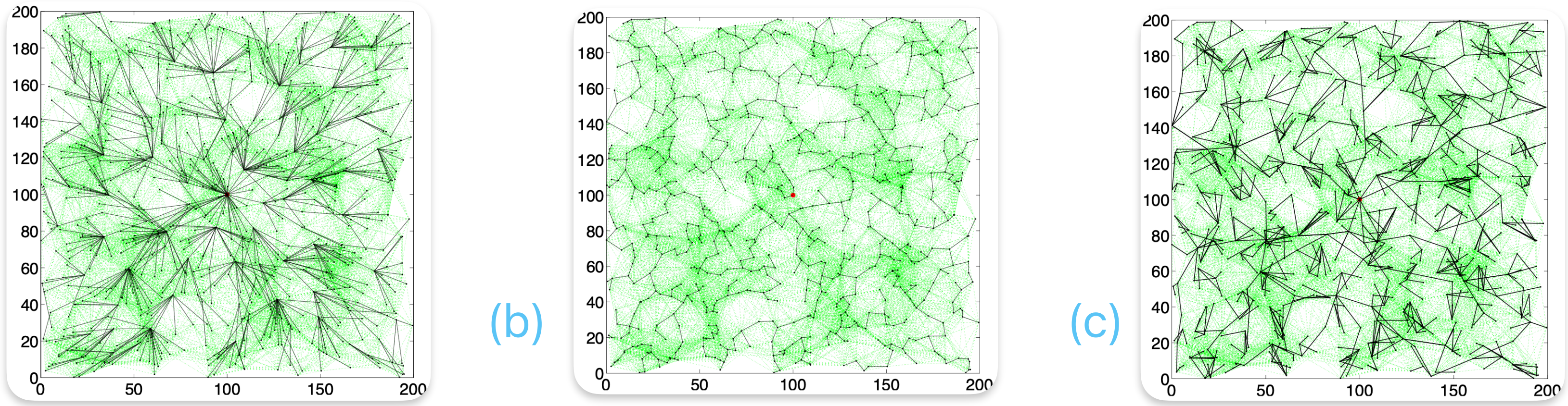

Fig 2. 800 nodes randomly deployed in a 200 x 200 region, with the sink at the center.
(a) Shortest-path tree: Minimum hop distances to the sink, but high node degrees,
(b) Minimum interference tree: Low node degrees, but large hop distances to the sink,
(c) BDMRST: Uniform node degrees, and uniform hop distances to the sink

Fig 1. Convergecast: Data flow from all nodes (a, ..., g) to sink s using a TDMA schedule. Arrows represent transmission directions, and numbers indicate time slots. A node transmits as per its receiver's frequency, e.g., d transmits over f1.

(Top) When transmissions occur over a single frequency f1, interfering links (dotted lines) come into play, and 5 time slots are needed for the sink to start receiving aggregated data from all the nodes in steady state.

(Bottom) When two frequencies f1 and f2 are used, interfering links disappear, and only 3 time slots are enough.